Data Mining

Data: Part 2

Mohammed Brahimi & Sami Belkacem

Exploratory Data Analysis (EDA)

- EDA is a set of **statistical** and **visualization** techniques.

- Used for seeing what the data can tell before the **preprocessing** and **modeling**:
  - Understand the data and summarize its keys properties.
  - Discover noisy data and outliers.
  - Comprehend the distribution of the data.
  - Decide which data cleaning techniques to be applied.

- EDA is cross-classified in two ways:
  - The method is either **non-graphical** or **graphical**.
  - The method is either **univariate** or **multivariate** (usually just bivariate).

2

Exploratory Data Analysis (EDA)

1. Statics of Data
2. Data visualization
3. How to choose an EDA technique

1. Statics of Data

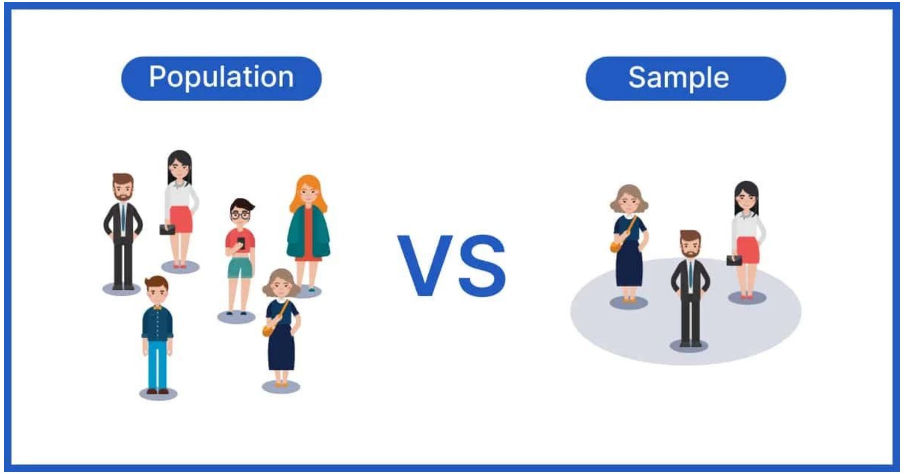
- Population vs Sample
- Measuring the Central Tendency (Mean, Median, and Mode)
- Symmetric vs. Skewed Data
- Measuring data distribution (Variance and Standard deviation)
- Analysis of two variables (Covariance and Correlation)

4

Population vs Sample

- **Population:** The entire group that we want to draw conclusions about.
- **Sample:** Subset of the population, used when the population size is too large.

It should be an unbiased subset that best represents the entire population.



Measuring the Central Tendency: (1) Mean

- Mean (algebraic measure) (sample vs. population):

  n is sample size and ***N*** is population size.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \mu = \frac{\sum x}{N}$$

- **Weighted arithmetic mean:**

$$\bar{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

- **Trimmed mean:** Chopping extreme values (e.g., olympics gymnastics score computation)

Measuring the Central Tendency: (2) Median

- **Median** is the middle value in a data set when values are ordered.

- **How to calculate**
  - Sort data in ascending order.
  - Repeat values according to their frequency.
  - Odd number of data points => the median is the middle value.
  - Even number of data points => the median is the average of the two middle values.

- **Why use the Median**
  - Resistant to extreme outliers compared to the mean.
  - Useful for skewed distributions.

7

Median calculation for grouped data

$$\text{Estimated Median} = L + \frac{(n/2) - B}{G} \times w$$

- **L:** lower class boundary of the median bin.
- **n:** total number of values.
- **B**: cumulative frequency of the bins before the median bin.
- **G**: frequency of the median bin.
- **W**: median bin width.

	Age	Frequency
0	5-10	4
1	10-20	5
2	20-50	3
3	50-55	1

Median calculation for grouped data

$$\textbf{Estimated Median} = L + \frac{(n/2) - B}{G} \times w$$

	Age	Frequency
0	5-10	4
1	10-20	5
2	20-50	3
3	50-55	1

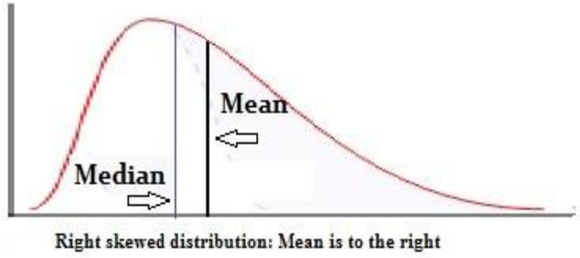
- **The median bin is the value 10-20**
  - L = 10
  - n = 13
  - B = 4
  - G = 5
  - w = 20-10 = 10

Median value = 10 + ((6.5-4)/5) *10 = 15

9

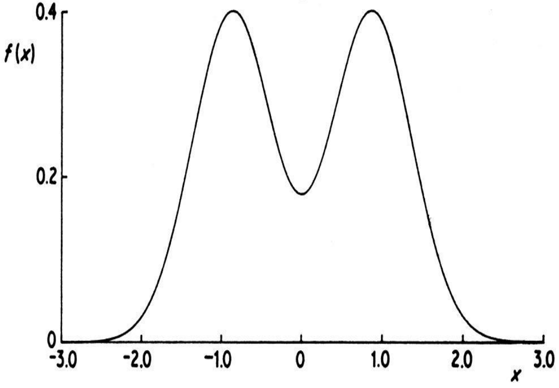
Measuring the Central Tendency: (3) Mode

- **Mode:** Value that occurs most frequently in the data

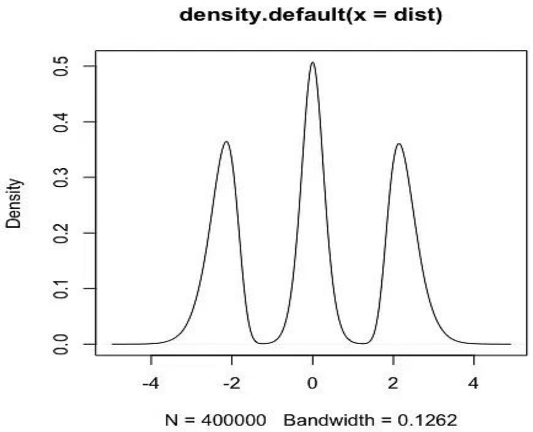


- **Unimodal**
  - Empirical formula: $mean - mode = 3 \times (mean - median)$

- **Multimodal**
  - Bimodal (a)
  - Trimodal (b)



(a)

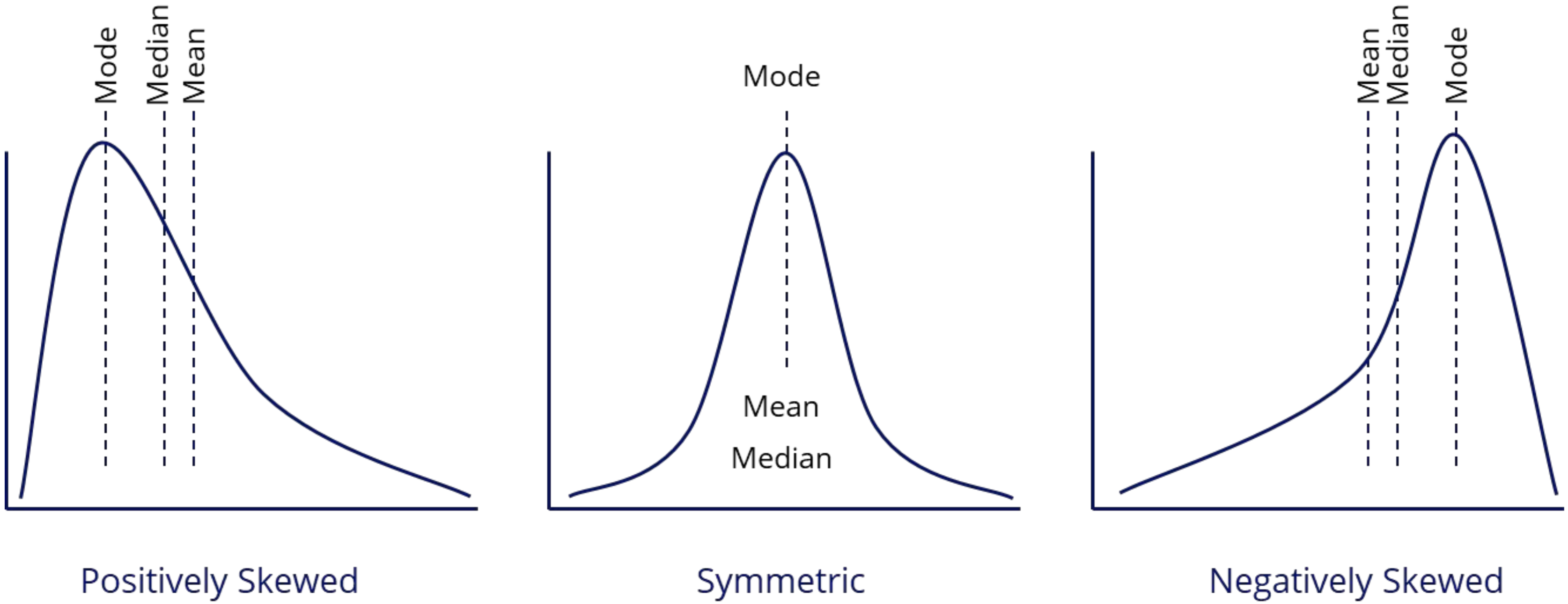


(b)

10

Symmetric vs. Skewed Data

- Median, mean, and mode of symmetric, positively, and negatively skewed data

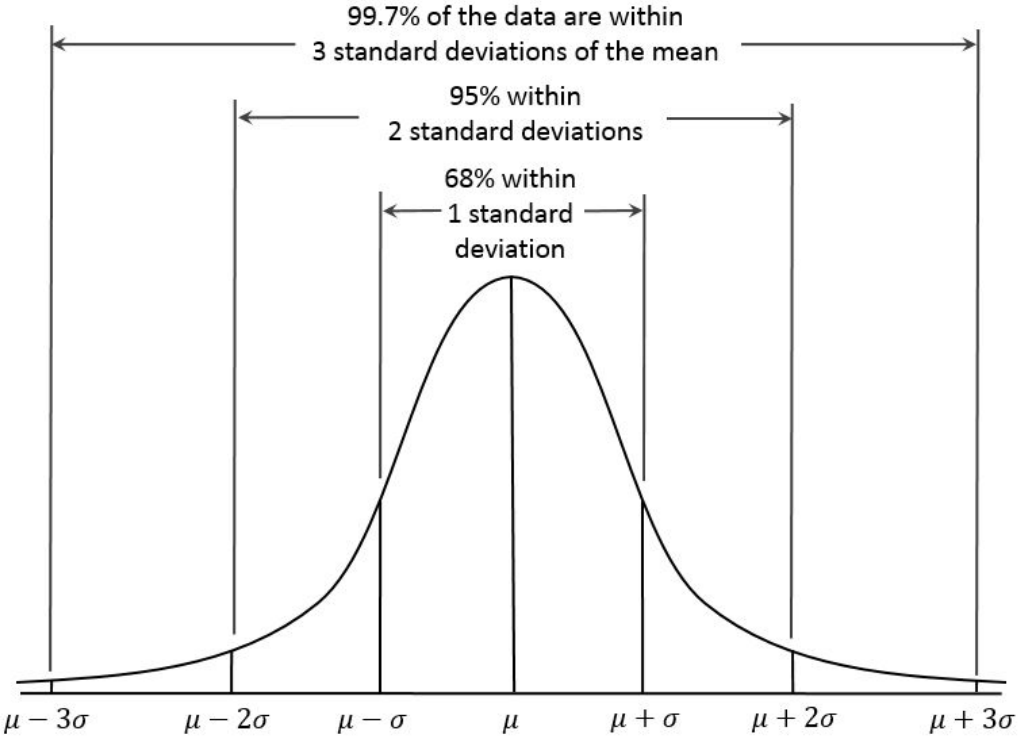


(Normal distribution)

11

Properties of Normal Distribution Curve

←— ————Represent data dispersion, spread — ————→

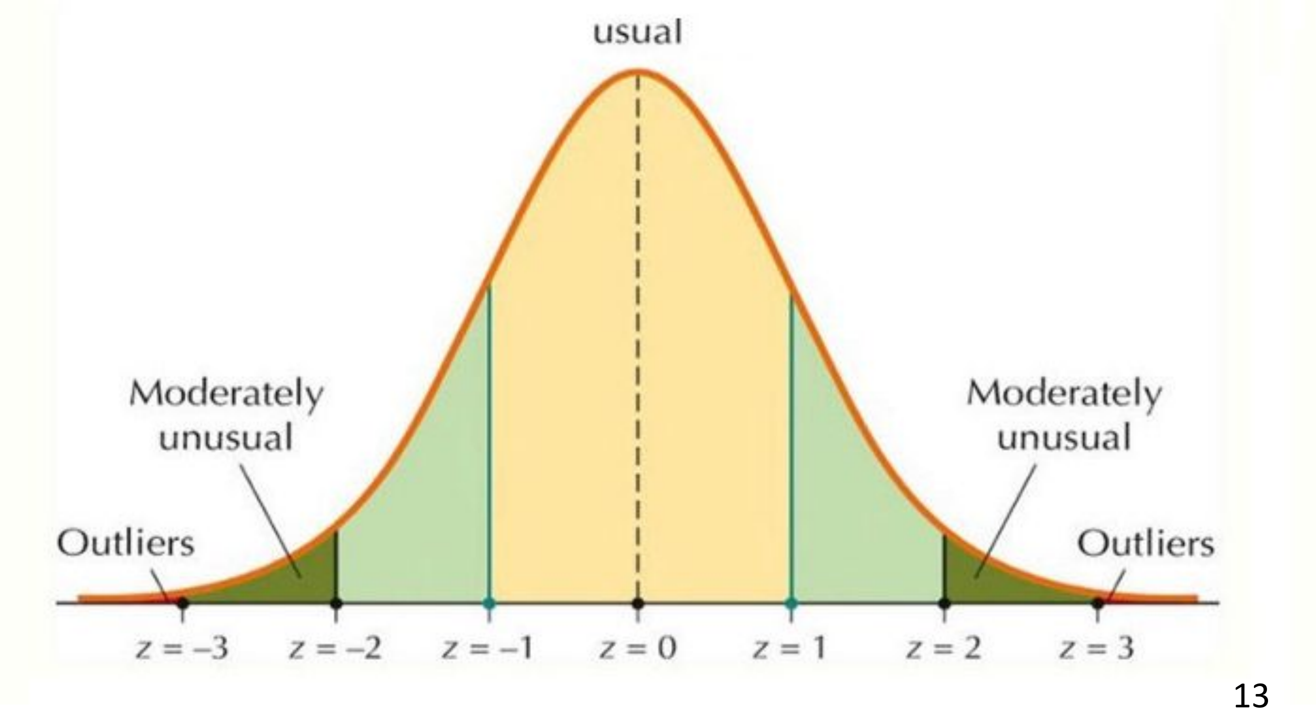


Represent central tendency

12

Properties of Data after z-score normalization

Z-score normalization (**μ**: mean, **σ**: standard deviation): $z = \frac{X - \mu}{\sigma}$



13

Measuring Data Distribution: Variance and Standard Deviation

- ***s:*** *sample*, ***σ:*** *population*, **n**: size of the sample, **N**: size of the population
  - **Variance** : Measure dispersion around the mean

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}[\sum_{i=1}^{n}x_i^{\ 2} - \frac{1}{n}(\sum_{i=1}^{n}x_i)^2]$$

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{n}(x_i - \mu)^2 = \frac{1}{N}\sum_{i=1}^{n}x_i^{\ 2} - \mu^2$$

  - For a **sample**: Divide by ***n−1*** to account for bias in the estimation.

    For a **population**: Divide by ***N*** because no correction is needed.

  - **Standard deviation** *s (or σ)* is the square root of variance s^2 *(or* σ^2*)*

    Measures dispersion around the mean, but in the same units as the value

Analysis of two variables: Covariance

$$cov_{x,y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

- The covariance for two numerical variables computes the relationship between them.
- Covariance shows how two variables change together:
  - **Positive Covariance:** both variables move together.
  - **Negative Covariance:** variables move in opposite directions.
  - **Zero Covariance:** no clear pattern in variable movements.
- Covariance is sensitive to scale, correlation addresses this by standardizing the measurement.

15

Covariance Matrix

- The variance and covariance information for the two variables can be summarized as 2X2 covariance matrix, which contains the following information:
  - The variance of each variable on the diagonal.
  - The covariance between the two variables in the off-diagonal entries.

$$\Sigma_{i,j} = \begin{bmatrix} \sigma_{ii}^2 & \sigma_{ij}^2 \\ \sigma_{ji}^2 & \sigma_{jj}^2 \end{bmatrix}$$

- Generalizing it to ***d*** dimensions: $\mathbf{\Sigma} = E[(\mathbf{X}-\mu)(\mathbf{X}-\mu)^T] = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{d1} & \sigma_{d2} & \cdots & \sigma_d^2 \end{pmatrix}$

16

Analysis of two variables: Correlation

- **Correlation** between two numerical variables $\mathbf{X_1}$ and $\mathbf{X_2}$ is the standard covariance, obtained by normalizing the covariance with the standard deviation of each variable.

$$\rho_{12} = \frac{\sigma_{12}}{\sigma_1 \sigma_2} = \frac{\sigma_{12}}{\sqrt{\sigma_1{}^2 \sigma_2{}^2}}$$

- **Sample correlation** for two attributes X_1 and X_2:

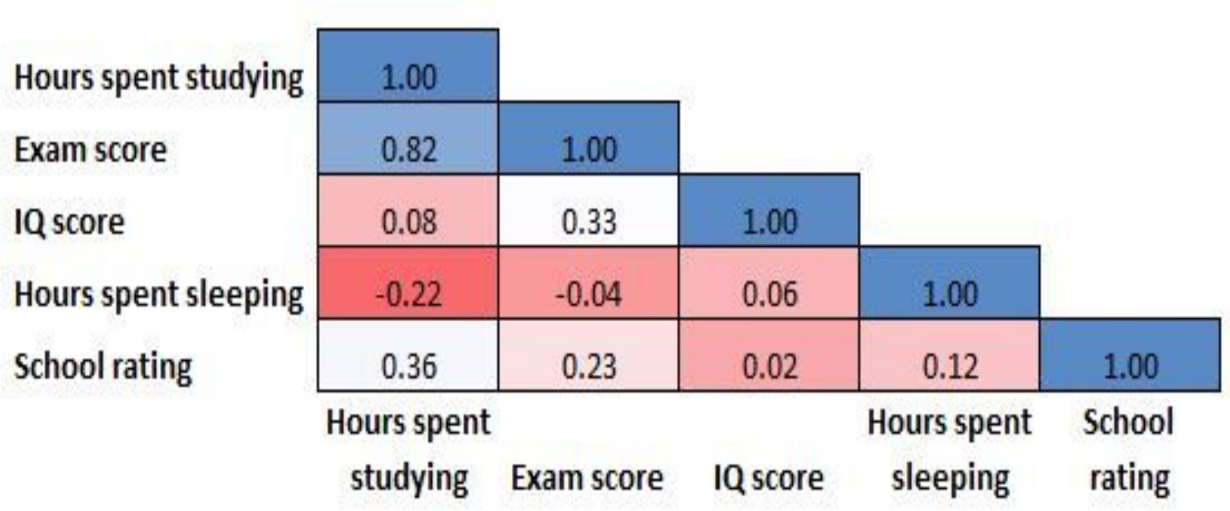
$$\hat{\rho}_{12} = \frac{\hat{\sigma}_{12}}{\hat{\sigma}_1 \hat{\sigma}_2} = \frac{\sum_{i=1}^{n}(x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2)}{\sqrt{\sum_{i=1}^{n}(x_{i1} - \hat{\mu}_1)^2 \sum_{i=1}^{n}(x_{i2} - \hat{\mu}_2)^2}}$$

n: number of tuples, $\boldsymbol{\mu_1}$ and $\boldsymbol{\mu_2}$: means of $\mathbf{X_1}$ and $\mathbf{X_2}$, $\boldsymbol{\sigma_1}$ and $\boldsymbol{\sigma_2}$: standard deviation of $\mathbf{X_1}$ and $\mathbf{X_2}$

- **If $\boldsymbol{\rho_{12} > 0}$:** A and B are positively correlated (X_1's values increase as X_2's)
- **If $\boldsymbol{\rho_{12} < 0}$:** A and B are negatively correlated (X_1's values increases, X_2's decreases)
- **If $\boldsymbol{\rho_{12} = 0}$:** independent (under the same assumption as discussed in co-variance)

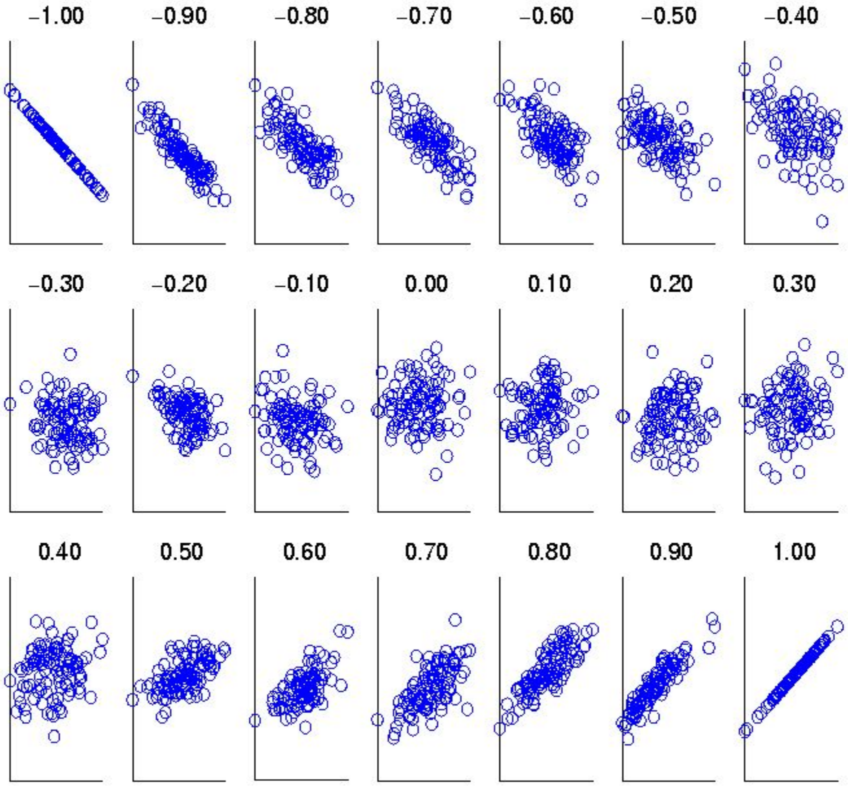
Correlation Matrix (Correlation Heatmap)

- Matrix that shows the correlations between each pair of variables in a dataset



18

Visualizing Changes of Correlation Coefficient



- Correlation coefficient value range: [–1, 1]
- A set of scatter plots shows sets of points and their correlation coefficients changing from –1 to 1

19

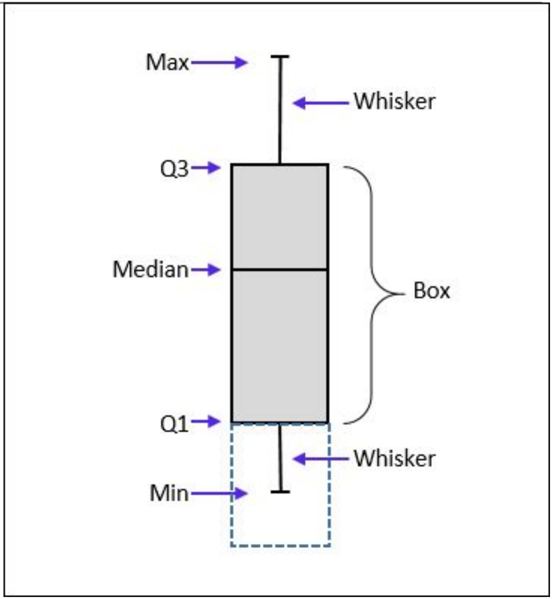
2. Data Visualization

- Boxplot
- Histogram and Bar chart
- Quantile plot
- Quantile-quantile (Q-Q) plot
- Scatter plot
- Line chart
- Parallel Coordinates plot

20

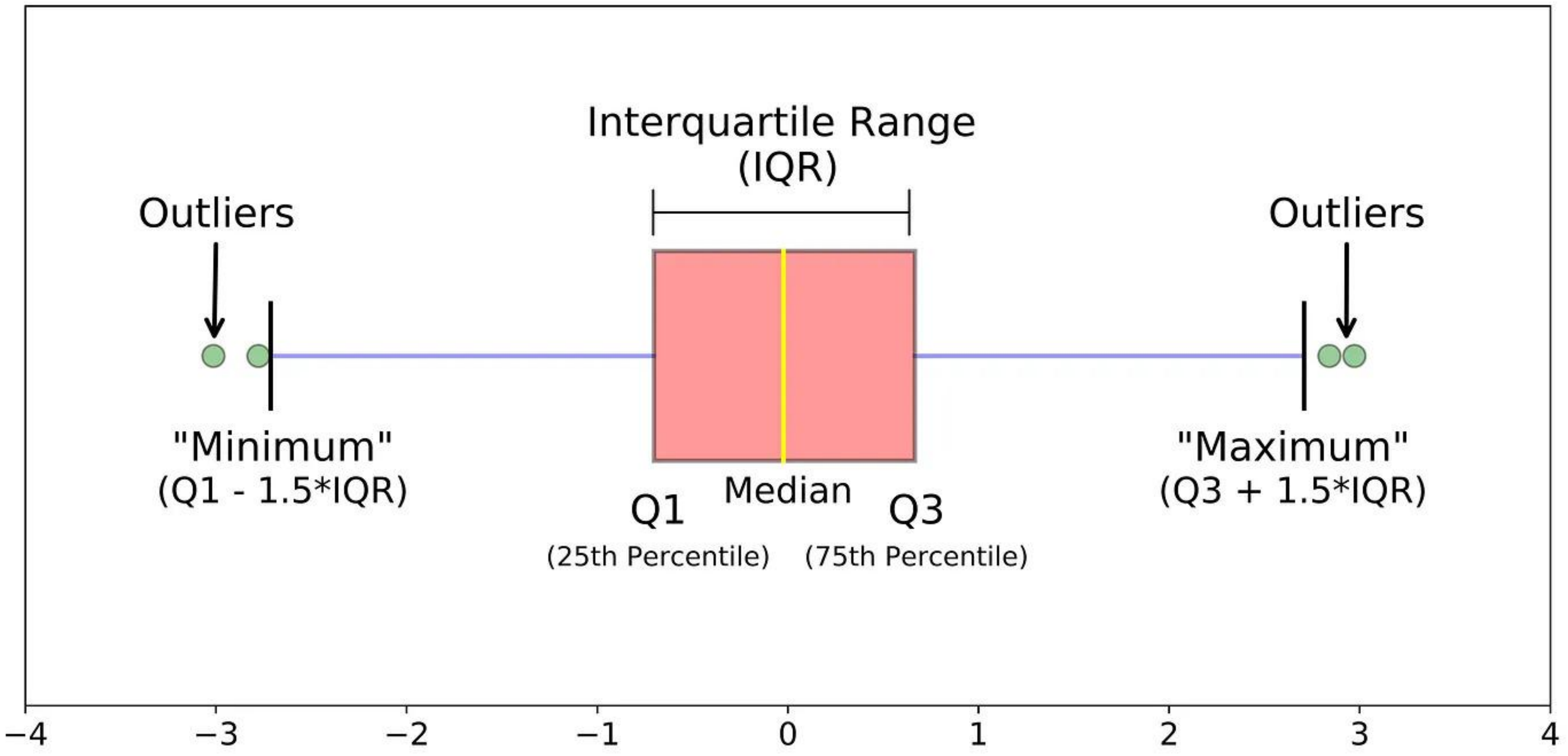
Boxplot: Data Dispersion

- **Quartiles**: Q_1 (25th percentile), Q_3 (75th percentile)
- **Interquartile range**: IQR = $Q_3 - Q_1$
- **Five number summary**: min, Q_1, median, Q_3, max
- **Box plot**: Data is represented with a box
  - **Q_1, Q_3, IQR**: The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
  - **Median (Q_2)** is marked by a line within the box
  - **Whiskers:** two lines outside the box extended to Minimum and Maximum
  - **Outliers:** points beyond a specified threshold (e.g. value higher/lower than 1.5 x IQR)



21

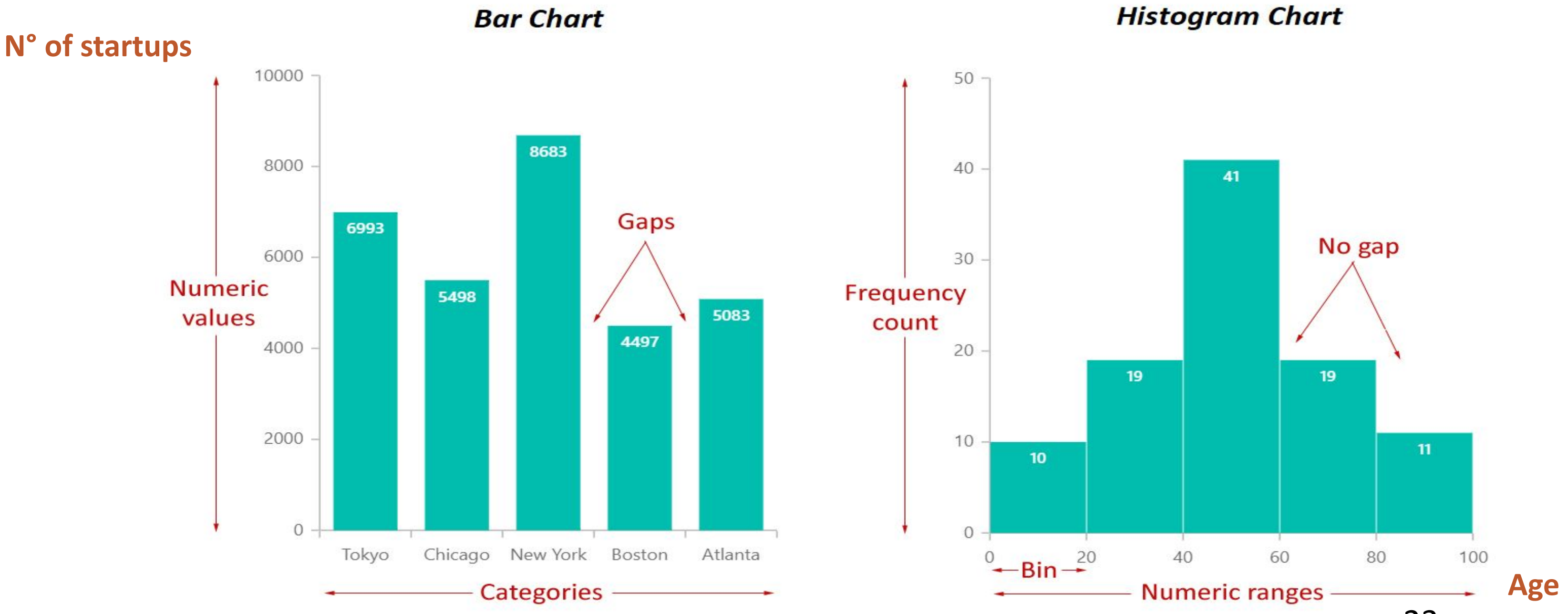
Boxplot: detect outliers



22

Histogram and Bar chart

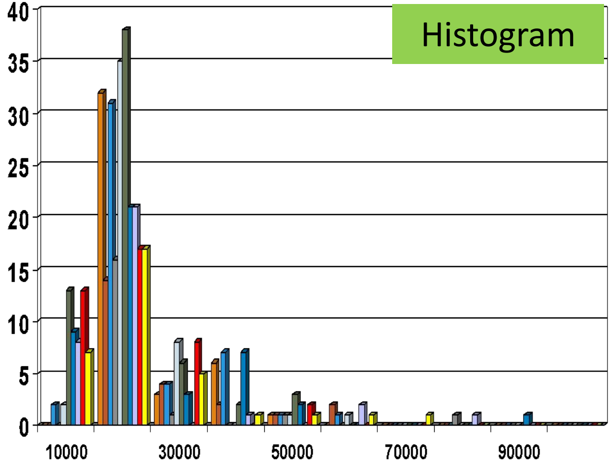
- **Histogram:** Tabulated frequencies represented by bars.
- **Bar chart:** Categorical data with bars proportional to the values they represent.

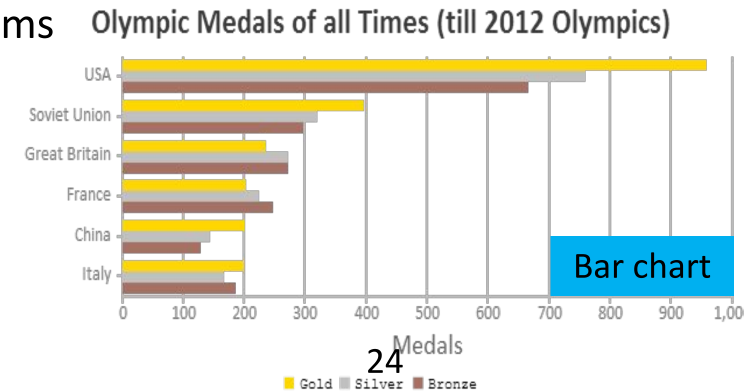


23

Histogram and Bar chart

- Histograms show **distributions** of variables, while bar charts compare **variables**
- Histograms plot binned **quantitative/categorical** data, while bar charts only plot **categorical** data
- Bars can be **reordered** in bar charts, but not in histograms
- In histograms, it is the area of the bar that denotes the value, not the height as in bar charts.

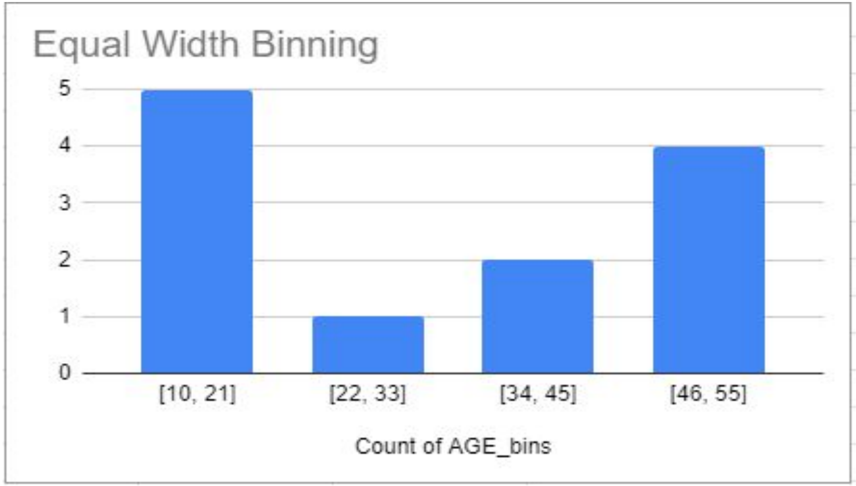


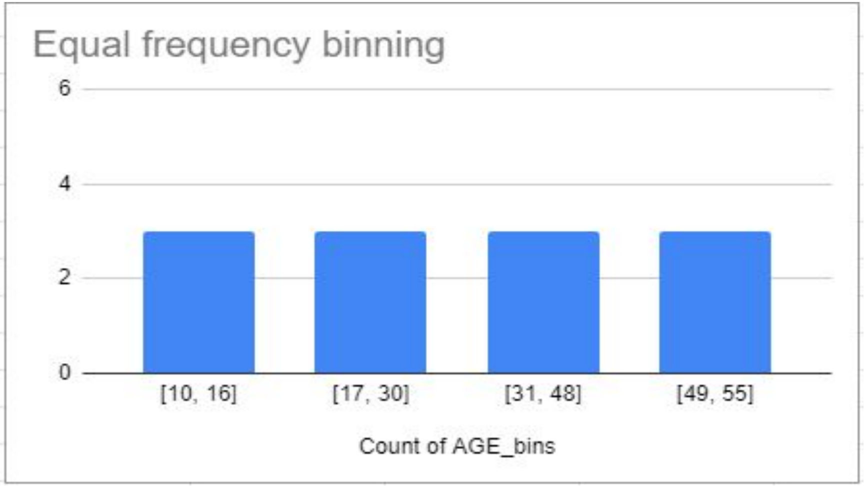


24

Histogram Analysis

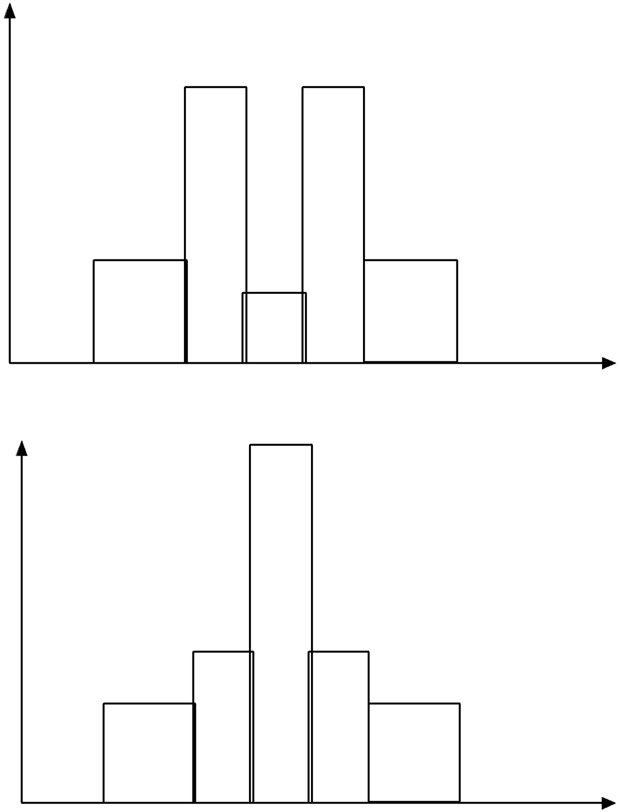
- Divide data into buckets and store average (sum) for each bucket.
- Partitioning rules:
  - **Equal-width:** equal bucket range
  - **Equal-frequency** (or equal-depth)





25

Histograms often tell more than Boxplots

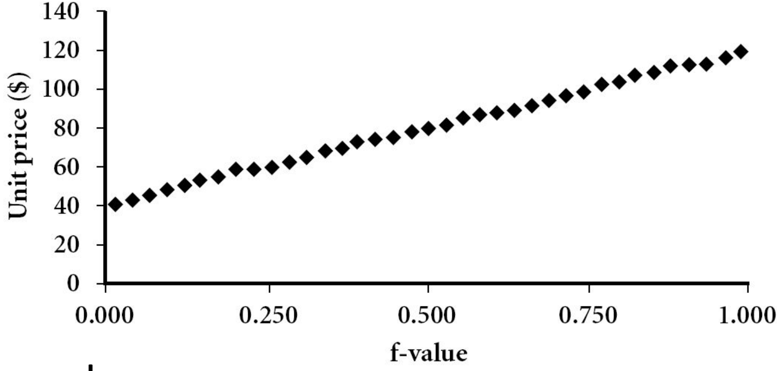


- The two histograms shown in the left may have the same box plot representation
  - The same values for: min, Q1, median, Q3, max
- But they have rather different data distributions

26

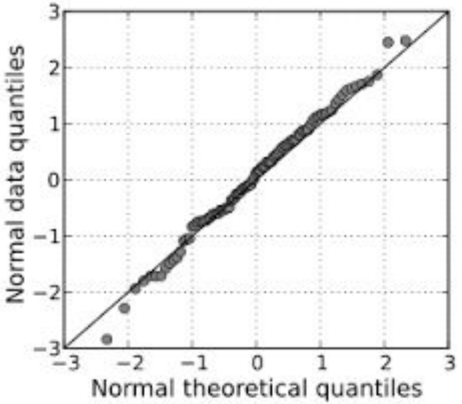
Quantile Plot

- **Purpose:** Visualizes all quantile information for a specific attribute.
- **Benefits:**
  - Provides a comprehensive view of the attribute's distribution
  - Helps identify general trends and outliers



- **Construction:**
  - For a data {X1, X2, …, XN} sorted in increasing order.
  - f_i indicates that approximately f_i of the data point have values ≤ x_i

Quantile-Quantile (Q-Q) Plot



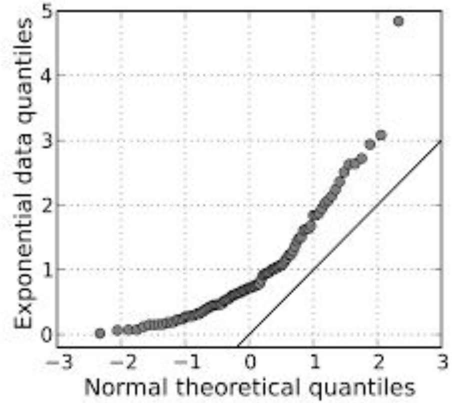
Definition:

Graphical tool used to compare the quantiles of two distributions.

- **X-axis:** Quantiles of the theoretical distribution
- **Y-axis:** Quantiles of the data distribution
- **Reference line:** A diagonal line (usually ***y***=***x***)

Purpose:

Assess the distributional similarity between an attribute and **another attribute** or a **theoretical distribution (e.g. normal distribution)**.



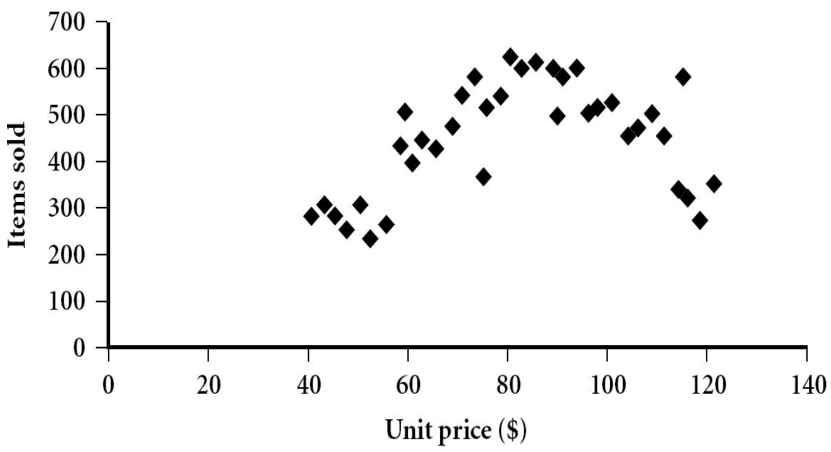
Interpretation:

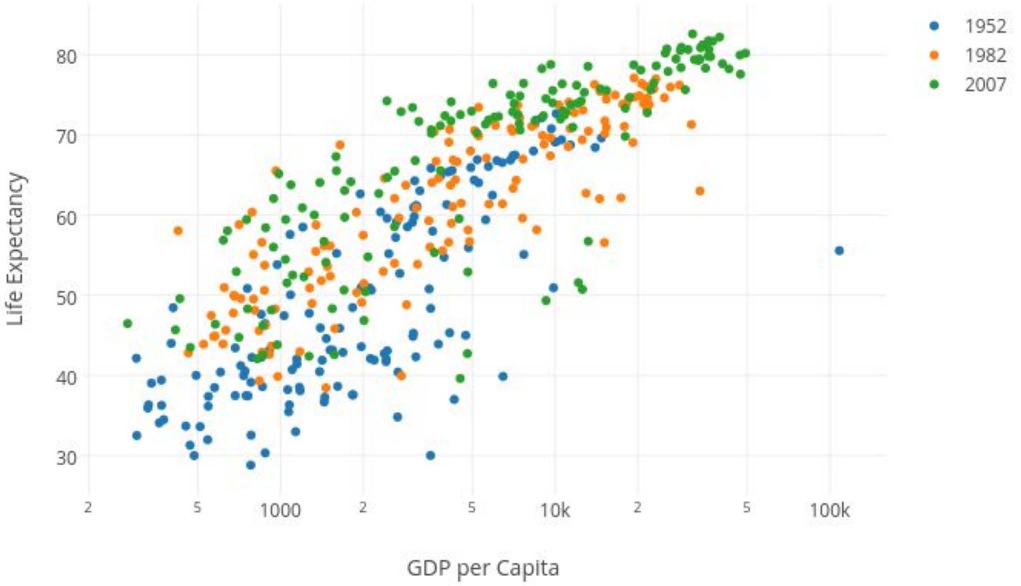
- **Close fit to straight Line**: The two distributions are similar
- **Deviations from the Line**: The two distributions are different

28

Scatter plot

- Provides a first look at bivariate data to see clusters of points, outliers, etc.
- Each pair of values is treated as a pair of coordinates and plotted as points in the plane.

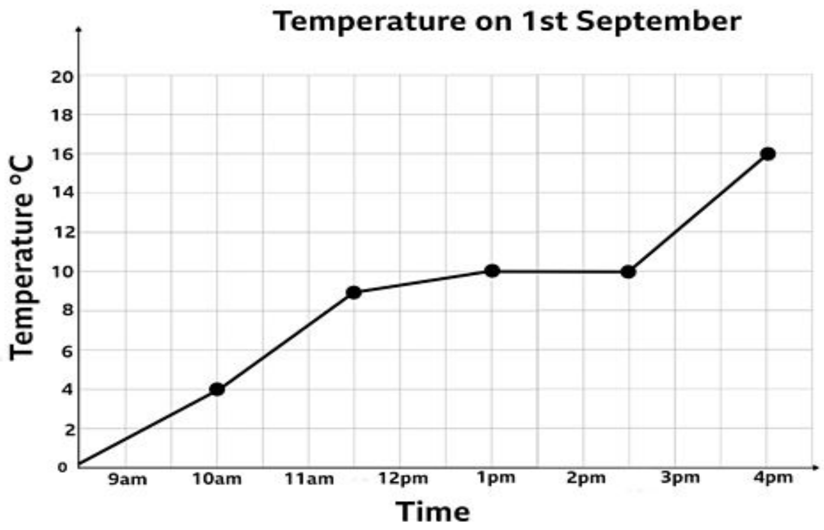




29

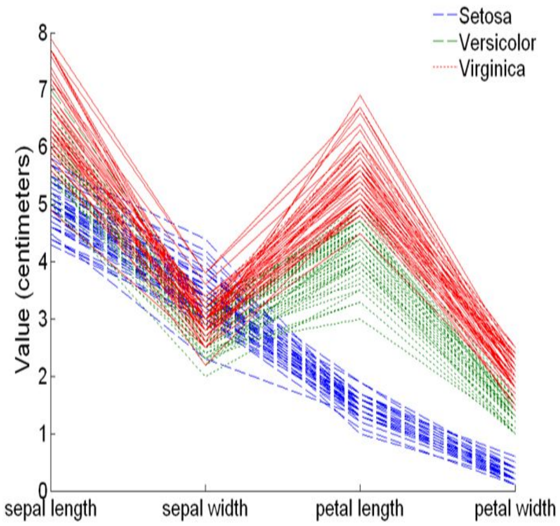
Line chart

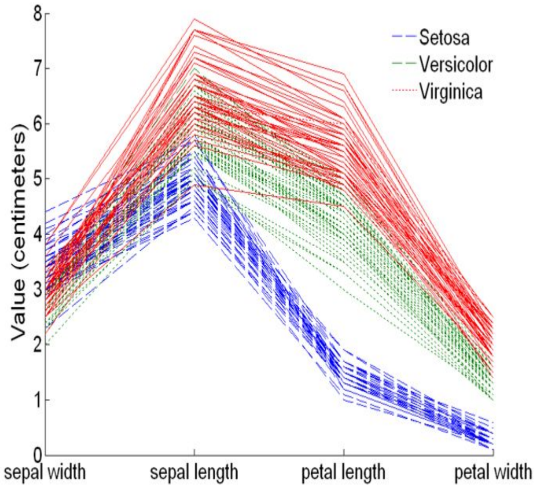
- A line chart displays information as a series of data points called 'markers'.
- The markers are connected to each other by straight line segments.

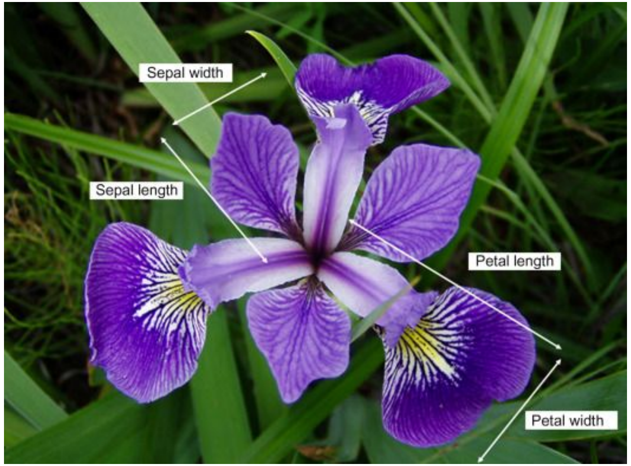


Parallel Coordinates Plots

- A parallel coordinate plot maps each object in the dataset as a line.
- Each attribute of an object is represented by a point on the line.
- The following example shows the parallel coordinates plots of the iris dataset.







31

3. How to choose an EDA technique

Criteria to choose a statistical/visualization technique:

- The type of analysis:
  - comparison, relationship, composition, distribution, dispersion, etc.
- The number of attributes (one, two, multiple attributes, etc.)
- The types of attributes (categorical, numerical, etc.)

The summary statistics: Summary statistics.pdf

The chart chooser: The chart chooser.pdf

Understanding Question

Given a dataset of objects, how to visually explore the relationship between two variables in the following cases?

- Continuous vs Continuous
- Categorical vs Continuous
- Continuous vs Categorical
- Categorical vs Categorical

33